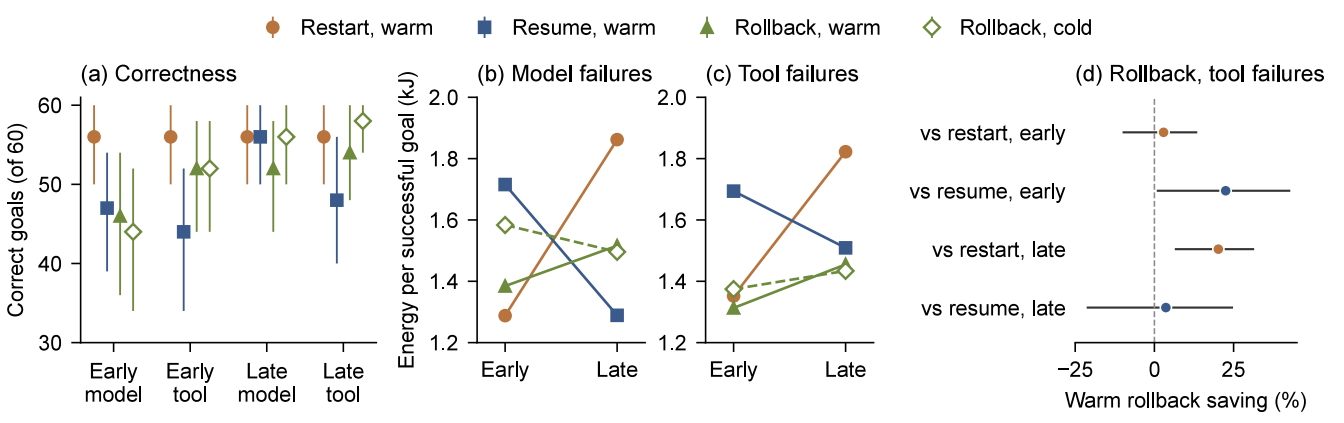

Restart, warm
Resume, warm
Rollback, warm
Rollback, cold
(a) Correctness
Correct goals (of 60)
60
50
40
30
Early model
Early tool
Late model
Late tool
(b) Model failures
Energy per successful goal (kJ)
2.0
1.8
1.6
1.4
1.2
Early
Late
(c) Tool failures
2.0
1.8
1.6
1.4
1.2
Early
Late
(d) Rollback, tool failures
vs restart, early
vs resume, early
vs restart, late
vs resume, late
−25
0
25
Warm rollback saving (%)